

Generation Interconnection System Impact Study Report

For Queue Project Q21-1843

Sydney Road 100 kV

January 24, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1843. The Interconnection Customer proposes a 82.3 MW battery energy storage facility interconnecting at the Sydney Road 100 kV substation in the Indiana area. The request was received on July 31, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 554 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,584,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: BONNET CRK – BUCHANAN MI 100 KV
Contingency: TABLE MT – GASTON RD SUBSTATION 100 KV Line
Loading: 135.7% (Rating: 137 MVA)
DFAX: 21.0%
Pre-project loading: 123.1%
2. Facility: BUCHANAN MI – NORTHERN SUBSTATION 100 KV
Contingency: GASTON RD – TABLE MT 100 KV
Loading: 134.3% (Rating: 138 MVA)
DFAX: 21.0%
Pre-project loading: 121.7%
3. Facility: AGER – NORTHERN 100 KV
Contingency: TABLE T – GASTON RD 100 KV
Loading: 130.3% (Rating: 162 MVA)
DFAX: 21.0%
Pre-project loading: 119.6%
4. Facility: TORTOISE – DENVER TERMINAL 100 KV Circuit 1
Contingency: TORTOISE – DENVER TERMINAL 100 KV
Loading: 117.6% (Rating: 239 MVA)
DFAX: 26.8%
Pre-project loading: 108.4%
5. Facility: TORTOISE – DENVER TERMINAL 100 KV Circuit 2
Contingency: TORTOISE – DENVER TERMINAL 100 KV Cir. 1
Loading: 117.6% (Rating: 239 MVA)
DFAX: 26.8%
Pre-project loading: 108.4%
6. Facility: TORTOISE – TEELAND 100 KV Cir. 1
Contingency: TEELAND – TORTOISE 100 KV Cir. 2
Loading: 115.1% (Rating: 265 MVA)

DFAX: 27.2%

Pre-project loading: 106.7%

7. Facility: TORTOISE – TEEAND 100 KV Ckt 2

Contingency: TEELAND – TORTOISE 100 KV L

Loading: 115.1% (Rating: 265 MVA)

DFAX: 27.2%

Pre-project loading: 106.7%

8. Facility: FREEHAVEN-LAKEPARK – TWIN BRANCH 100 KV L

Contingency: TWIN BRANCH PV – PARKROSE 100 KV

Loading: 113.7% (Rating: 300 MVA)

DFAX: 34.7%

Pre-project loading: 104.1%

9. Facility: TEELAND – TABLE MT 100 KV Ckt 1

Contingency: TEELAND – TALE MT 100 KV

Loading: 113.1% (Rating: 300 MVA)

DFAX: 28.4%

Pre-project loading: 105.3%

10. Facility: TEELAND – TABLE MT 100 KV Line 2

Contingency: TEELAND – TABLE MT 100 KV Circuit 1

Loading: 113.1% (Rating: 300 MVA)

DFAX: 28.4%

Pre-project loading: 105.3%

11. Facility: TABLE MT – GASTON RD 100 KV

Contingency: KEMPER COUNTY – DALREED 100 KV

Loading: 112.7% (Rating: 408 MVA)

DFAX: 67.0%

Pre-project loading: 99.2%

12. Facility: TWIN BRANCH – KEMPER COUNTY 100 KV

Contingency: PARKOSE – TWIN BRANCH PV 100 KV

Loading: 112.6% (Rating: 329 MVA)

DFAX: 34.7%

Pre-project loading: 104.0%

13. Facility: DENVER TERMINAL – FLOOD ST 100 KV #1

Contingency: NORTH MESQUITE – DENVER TERMINAL 100 KV

Loading: 106.4% (Rating: 146 MVA)

DFAX: 11.4%

Pre-project loading: 100.0%

14. Facility: DENVER TERMINAL – FLOOD STREET 100 KV

Contingency: NORTH MESQUITE – DENVER TERMINAL 100 KV

Loading: 106.4% (Rating: 146 MVA)

DFAX: 11.4%

Pre-project loading: 100.0%

15. Facility: NORTH MESQUITE – DENVER TERMINAL 100 KV Line

Contingency: TWIN BRANCH PV – PARKROSE 100 KV L

Loading: 102.2% (Rating: 225 MVA)

DFAX: 12.7%

Pre-project loading: 97.5%

16. Facility: SHARON SPRING – SYDNEY ROAD SUBSTATION 100 KV

Contingency: FORT HUMBURG SUBSTATION – ORCHARD RD 500 KV

Loading: 101.7% (Rating: 125 MVA)

DFAX: 100.0%

Pre-project loading: 36.1%

17. Facility: TWIN BRANCH PV – PARKROSE 100 KV Line
Contingency: TWIN BRANCH – KEMPER COUNTY 100 KV
Loading: 100.4% (Rating: 356 MVA)
DFAX: 21.9%
Pre-project loading: 95.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BUCHANAN MI – BONNET CREEK 100 KV L	Rebuild line with higher-capacity conductor	\$2,622,000	9% (\$242,000)
BUCHANAN MI – NORTHERN SUBSTATION 100 KV	Rebuild line with higher-capacity conductor	\$5,812,000	9% (\$536,000)
AGER – NORTHERN 100 KV	Rebuild line with higher-capacity conductor	\$3,623,000	9% (\$334,000)
TORTOISE – DENVER TERMINAL 100 KV Circuit 1	Rebuild line with higher-capacity conductor	\$2,484,000	6% (\$137,000)
TORTOISE – DENVER TERMINAL 100 KV Circuit 2	Rebuild line with higher-capacity conductor	\$2,186,000	6% (\$121,000)
TORTOISE – TEELAND 100 KV Cir. 1	Rebuild line with higher-capacity conductor	\$1,845,000	6% (\$102,000)
TORTOISE – TEEAND 100 KV Ckt 2	Rebuild line with higher-capacity conductor	\$1,887,000	6% (\$104,000)
TWIN BRANCH – FREEHAVEN-LAKEPARK 100 KV	Rebuild line with higher-capacity conductor	\$2,845,000	7% (\$190,000)
TEELAND – TABLE MT 100 KV Ckt 1	Rebuild line with higher-capacity conductor	\$1,017,000	5% (\$55,000)
TEELAND – TABLE MT 100 KV Line 2	Rebuild line with higher-capacity conductor	\$925,000	5% (\$50,000)
GASTON RD – TABLE MT 100 KV	Rebuild line with higher-capacity conductor	\$764,000	12% (\$88,000)
KEMPER COUNTY SUBSTATION – TWIN BRANCH 100 KV	Rebuild line with higher-capacity conductor	\$3,010,000	7% (\$201,000)
DENVER TERMINAL – FLOOD ST 100 KV #1	Reconductor line	\$1,032,000	5% (\$50,000)
DENVER TERMINAL – FLOOD STREET 100 KV	Reconductor line	\$2,457,000	5% (\$119,000)
NORTH MESQUITE – DENVER TERMINAL 100 KV Line	Reconductor line	\$913,000	4% (\$39,000)
SHARON SPRING – SYDNEY ROAD SUBSTATION 100 KV	Reconductor line	\$2,170,000	100% (\$2,170,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TWIN BRANCH PV – PARKROSE 100 KV Line	Reconductor line	\$1,082,000	4% (\$46,000)
	Total allocated to project		\$4,584,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.